

1. List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

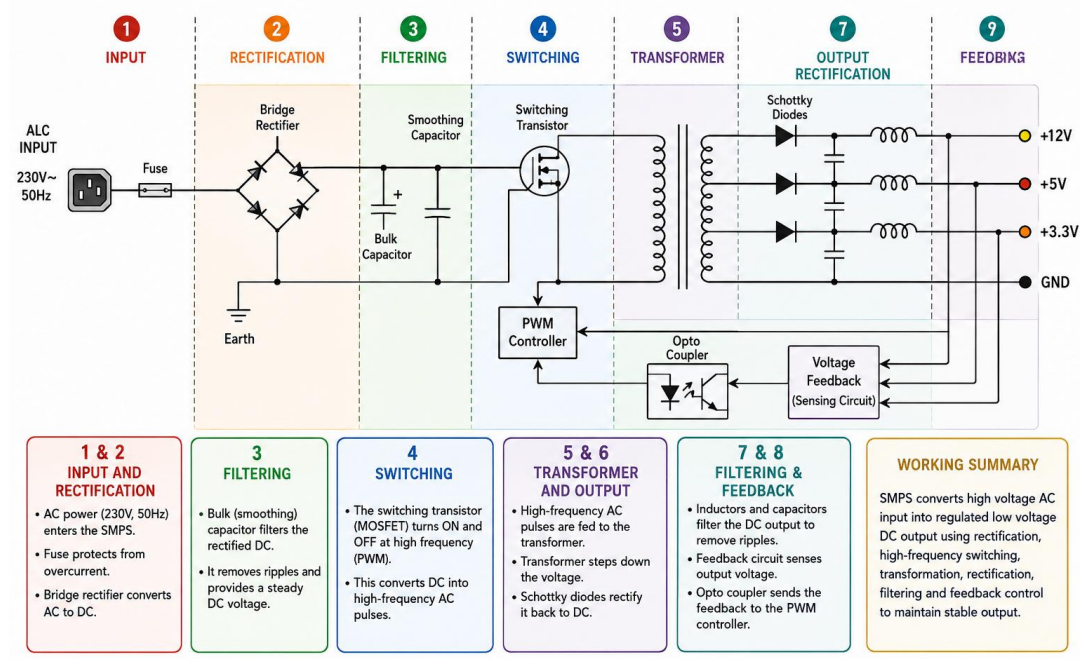
ANS:- Real-World Devices Using SMPS and UPS

Device	Type	Why it needs this power supply
Desktop Computer (Home/College Lab)	SMPS	Converts 230V AC into stable DC voltages (+12V, +5V, +3.3V) for computer components.
Laptop Charger	SMPS	Efficiently converts AC power into low-voltage DC required to charge the battery.
LED Monitor	SMPS	Provides regulated DC power while minimizing energy loss and heat.
Desktop Computer Backup	UPS	Keeps the PC running during power cuts, preventing data loss.
Wi-Fi Router Backup	UPS	Maintains internet connectivity during power outages.

2. Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections. Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io.

ANS:-

TYPICAL SMPS (SWITCH MODE POWER SUPPLY) – INTERNAL BLOCK DIAGRAM



3. Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

ANS:- Power Connectors Used in a Desktop PC

1. ATX 24-pin Motherboard Connector

Image:-



Devices it Powers:

- Motherboard (main power supply)
- CPU, RAM, PCIe slots, and onboard components receive power through the motherboard.

Safety Tip:

Always switch off the SMPS and unplug the AC power cable before connecting or disconnecting the 24-pin connector. Ensure it clicks into place securely.

2. SATA Power Connector

Image:



Devices it Powers:

- SATA Hard Disk Drives (HDDs)
- SATA Solid State Drives (SSDs)
- DVD/Blu-ray optical drives

Safety Tip:

Insert the connector in the correct orientation using the L-shaped key. Never force the connector, as it can damage the device.

3. Molex 4-pin Power Connector

Image:



Devices it Powers:

- Older IDE hard drives
- Optical drives
- Case fans
- Some LED accessories and cooling devices

Safety Tip:

Hold the plastic connector while unplugging. Do not pull on the wires, as they may become loose or damaged.

4. PCIe 6/8-pin Power Connector

Image:



Devices it Powers:

- Dedicated Graphics Cards (GPUs)
- High-performance PCIe expansion cards

Safety Tip:

Make sure the connector is fully inserted and the locking clip is engaged before turning on the computer. Never force a connector that does not match the GPU socket.

4. Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

ANS:- Steps to Connect the SMPS in a New Gaming PC

1. **Turn off the power supply** and unplug it from the electrical outlet.
2. **Connect the 24-pin ATX cable** from the SMPS to the **motherboard's 24-pin ATX power socket**.
3. **Connect the CPU power cable (4+4-pin or 8-pin EPS connector)** to the CPU power socket near the processor.
4. **Install the graphics card (GPU)** into the PCIe x16 slot on the motherboard.
5. **Connect the PCIe 6-pin or 8-pin power cable** from the SMPS to the graphics card.
6. **Connect SATA power connectors** to all SSDs, HDDs, and SATA optical drives.
7. **Connect SATA data cables** from the motherboard to each storage device (these carry data, not power).
8. **Connect Molex connectors** only if older fans or accessories require them.
9. **Check that every connector is fully inserted** and locked into place.
10. **Reconnect AC power**, switch on the SMPS, and start the PC to verify that all components are powered correctly.